

*"Design it, build it, secure it, ship it — then
let AI help, under control."*

COURSE INTRODUCTION

Spring Boot Microservices with OpenShift, AI and DevOps

Five days, twelve modules and a capstone — one banking platform, built lab by lab.

Day 1
Design and build

Day 2
Integrate and data

Day 3
Secure and harden

Day 4
Deliver on OpenShift

Day 5
AI, MCP and capstone



WELCOME TO THE COURSE

A warm welcome, a few ground rules and what the week looks like.



WELCOME

- We're glad you're here
- Five days of learning, building and sharing ideas
- Practical skills you can use with your team right away



IN THIS COURSE, YOU WILL

- Learn by building one real banking platform
- Work through realistic failure and security scenarios
- Collaborate and share ideas with your peers
- Get guidance and feedback on your lab work



GROUND RULES

BE OPEN

Ask questions early — even the ones that feel basic.

BE RESPECTFUL

Every experience level in the room adds something.

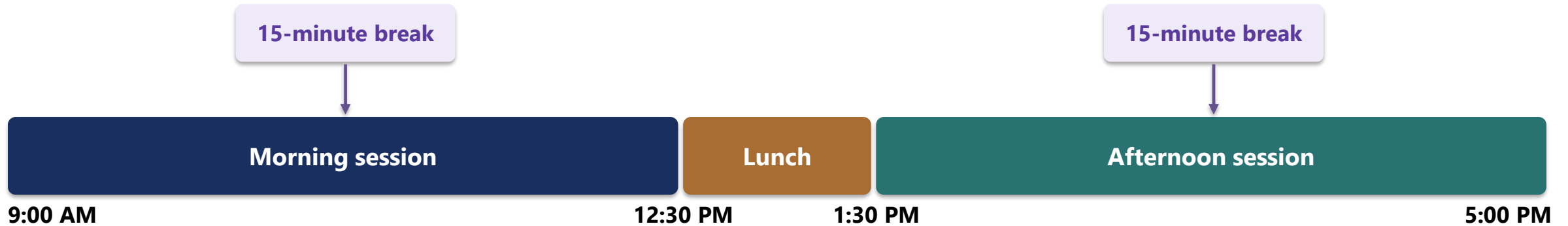
BE ENGAGED

Try every lab step yourself, not just watch.

KEY TAKEAWAY Five days of hands-on learning — ask early, try everything, share ideas.

LOGISTICS

Daily schedule, breaks and a few housekeeping notes.



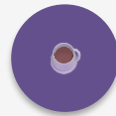
Class hours

9:00 AM – 5:00 PM EST, all five days.



Lunch

One hour, around 12:30 PM.



Breaks

15 minutes in the morning and 15 minutes in the afternoon.



Phones

Please keep your phone on silent.

KEY TAKEAWAY 9 AM to 5 PM EST, lunch around 12:30, two 15-minute breaks — phones on silent.

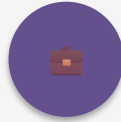
LET'S GET TO KNOW EACH OTHER

Take a minute to introduce yourself to the group.



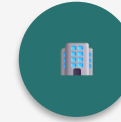
Your name

What should we call you?



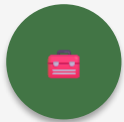
Your role

What do you do day to day?



Your team

Which team or line of business are you joining from?



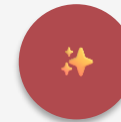
Your experience

Spring Boot, microservices, containers, CI/CD or AI tools?



Your goal

What do you hope to take back to your team?



Something about you

A hobby, a fun fact or a favourite place.

HOW IT WORKS About one minute each. The instructor goes first.

KEY TAKEAWAY Knowing the room helps us connect the material to your real work.

THE COURSE AT A GLANCE

What's in the course — and what you'll leave with.

5

days

9 AM – 5 PM EST

12 + 1

modules + capstone

One connected story

6

hands-on labs

Lab 0 setup + Labs 1–5

15

checkpoints

Three official exercises a day

3

banking services

Account · Transaction · Risk



COURSE HIGHLIGHTS

- Microservice design, REST APIs and Spring Boot
- Events, data ownership and resilience patterns
- OAuth2, JWT scopes, secrets and audit
- Containers, OpenShift and CI/CD pipelines
- GitHub Copilot, agentic AI and MCP — governed



BY FRIDAY, YOU WILL HAVE

- Three services with their own data
- Secured, resilient and tested APIs
- Services deployed to OpenShift through a pipeline
- An AI risk score with policy and human review
- A capstone demo that proves it all works

KEY TAKEAWAY Twelve modules, six labs and a capstone — all building one working platform.

WHAT YOU'LL BUILD

The AI-Assisted Banking Transaction Risk Platform — one layer each day.



ONE PLATFORM, FIVE LAYERS



ACCOUNT SERVICE · LAB 1

Account lifecycle: create, activate, freeze, close. REST, validation, OpenAPI and Actuator.

TRANSACTION SERVICE · LAB 2

Checks the account over REST and publishes TransactionSubmitted events.

RISK ASSESSMENT SERVICE · LAB 5

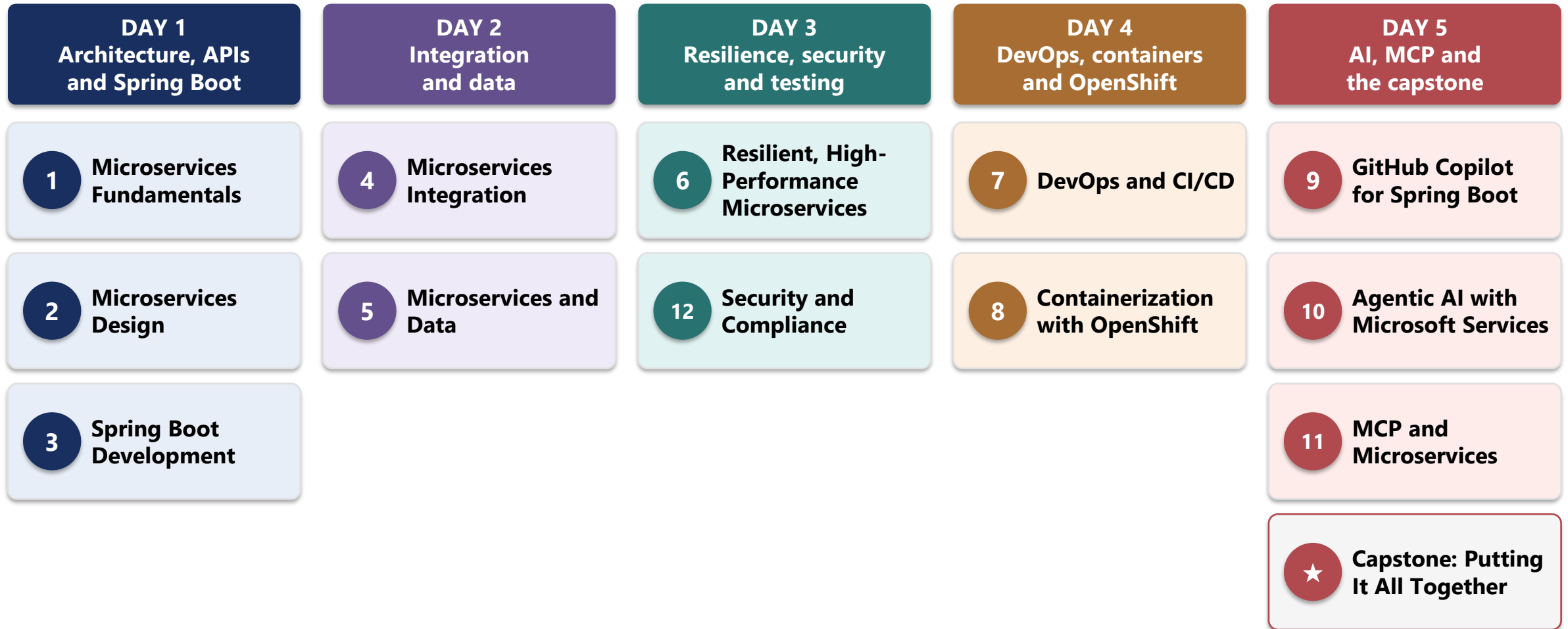
An OpenShift AI model scores; Java policy decides; people review every HOLD.

CUMULATIVE Each lab builds on the one before. Starters and solutions keep everyone moving.

KEY TAKEAWAY Labs aren't separate exercises — together they assemble one three-service platform.

TWELVE MODULES AND A CAPSTONE

Grouped by the day we teach them.



KEY TAKEAWAY Module 12 (Security) is taught on Day 3, so Lab 3 can secure the services.

WHAT WE COVER IN FIVE DAYS

Topics, labs and checkpoint exercises for each day.

DAY 1

Architecture, API design and Spring Boot

Microservice fundamentals, service boundaries, REST design, Spring Boot layers, validation, OpenAPI and Actuator.

Lab 0 Environment Setup
Lab 1 Account Service
Exercises 1.1–1.3

DAY 2

Integration, data and events

Sync vs. async calls, Kafka events, idempotency and DLQs, database per service, JPA, Flyway and transactions.

Lab 2 Transaction Service
Exercises 2.1–2.3

DAY 3

Resilience, security and testing

Timeouts, retries, circuit breakers, observability, OAuth2 and OIDC, JWT scopes, secrets and security tests.

Lab 3 Secure and Resilient Services
Exercises 3.1–3.3

DAY 4

DevOps, containers and OpenShift

Git and PR gates, CI/CD, image scanning, containers, Deployments, Services, Routes, probes and rollouts.

Lab 4 Deploy the Capstone Services
Exercises 4.1–4.3

DAY 5

OpenShift AI, MCP and the capstone

GitHub Copilot, agentic AI, model calls with safe fallbacks, policy and human review, and MCP.

Lab 5 Risk Assessment Service
Exercises 5.1–5.3 · Capstone demo

KEY TAKEAWAY Every day follows the same pattern: modules, checkpoints, then a lab.

HOW EACH DAY RUNS

Short concept blocks, practice and checkpoints — then build it in the lab.



THE RHYTHM OF EACH MODULE



THEN THE LAB

Each day ends with a hands-on lab that applies the day's modules to the banking platform.

NEVER STUCK

Every lab has a starter and a solution, so a missed step never blocks the next day.

ASK ANY TIME

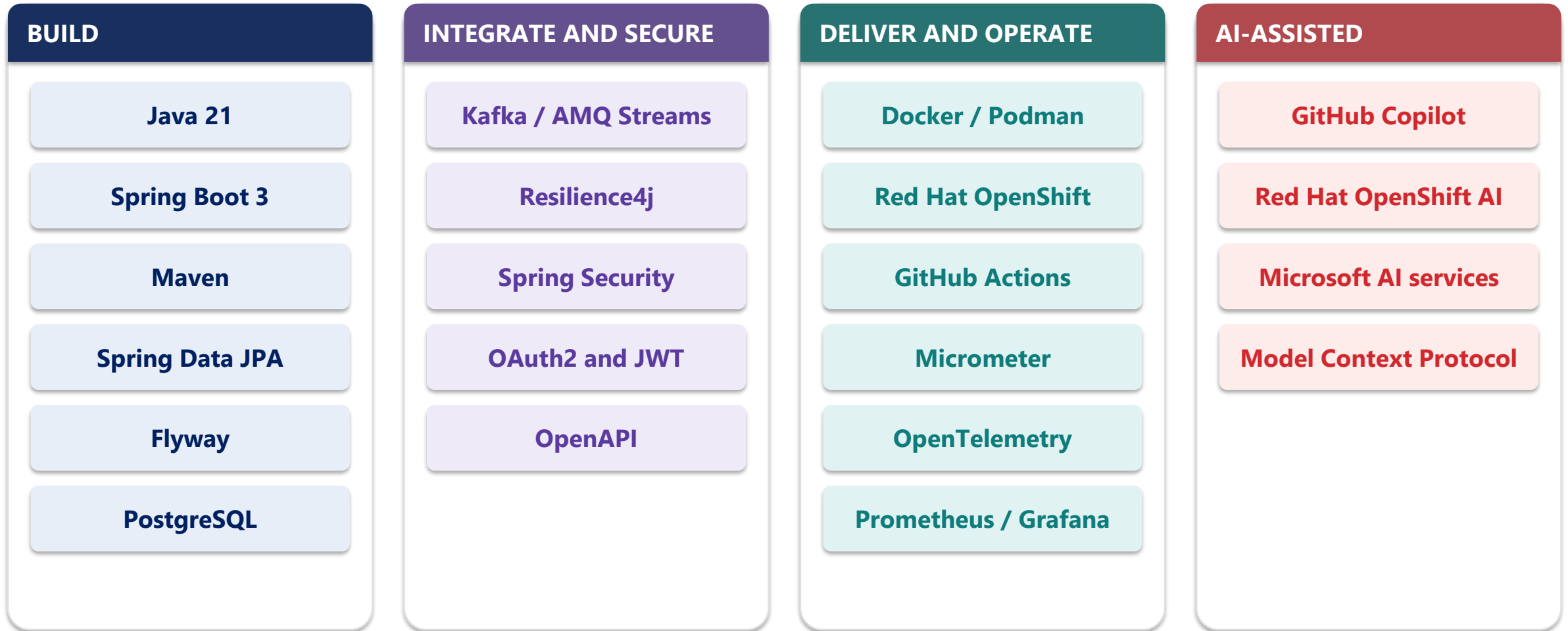
Questions in the room or in chat are always welcome — the lab time is for trying things.

LOW STAKES Quizzes and knowledge checks are for you — a quick way to see what's clicked.

KEY TAKEAWAY Learn a concept, practise it, check it, then build it.

TECHNOLOGY STACK

The tools and platforms we use across the week.



KEY TAKEAWAY Enterprise-standard tools, used the way production teams use them.

"Ask early — every question helps the whole room."

LET'S GET STARTED

Questions Before We Begin?

Next: the Lab 0 environment check, then Module 1 — Microservices Fundamentals.